

NAINA OS -- VOLUME 4

MEMORY & KNOWLEDGE INTELLIGENCE ENGINE (MKIE v1.0)

"A model answers questions. A memory system builds a mind."

DOCUMENT METADATA

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Core Focus:	7-Layer Memory * 16-Entity Graph * Obsidian Vault Sync

This document contains formal engineering specifications for the NAINA OS Unified Memory Engine, 16-Entity Knowledge Graph, Obsidian Vault Integration, Personal Knowledge Timeline, and Identity Engine.

NAINA OS -- Memory & Knowledge Intelligence Engine (MKIE v1.0)

Volume 4: Multi-Tier Memory Engine, Knowledge Graph, Obsidian Integration & Identity Engine

A model answers questions. A memory system builds a mind.

Document Revision History

Date	Revision	Author	Description of Changes
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2026-07-31	v1.0.0	Chief Systems Architect	Release of Volume 4 -- Memory & Kno
2026-07-26	v0.9.0	Memory Engineering Team	Specifications for 7 Memory Layers,

DOCUMENT 04.1 -- Unified Memory Engine (UME)

1. Overview & Multi-Domain Memory Concept

The Unified Memory Engine (UME) provides a single, cross-modal memory system servicing all user interaction domains: Voice, Chat, Vision, Desktop, Android, Browser, Research, Coding, and Photography.

```
+-----+
|               UNIFIED MEMORY ENGINE (UME) LAYERS               |
+-----+
| LAYER 1: SENSORY MEMORY (Microphone Audio Buffer, Frame Captures) |
+-----+
| LAYER 2: WORKING MEMORY (Active Task State: Docker -> GitHub -> Railway)|
+-----+
| LAYER 3: SESSION MEMORY (Daily Operations: Coding, Meetings, Study) |
+-----+
| LAYER 4: PROJECT MEMORY (Academic Hub, NAINA OS, Photography, YouTube) |
+-----+
| LAYER 5: LONG-TERM MEMORY (Goals, Habits, Coding Style, Life History) |
+-----+
| LAYER 6: KNOWLEDGE GRAPH (16 Entity Types, Triples & Relations)      |
+-----+
| LAYER 7: OBSIDIAN VAULT (Intelligent Plaintext Markdown .md Mirror)  |
+-----+
```

2. Detailed Memory Layer Specifications

- Sensory Memory:
 - * ***Lifespan*:** 500ms - 10 seconds (In-RAM ring buffer).
 - * ***Examples*:** Raw PCM microphone stream, latest camera frame, clipboard data, active screen viewport bounding box.
- Working Memory:
 - * ***Lifespan*:** Active task duration (Minutes to hours).

- * *Examples*: Active goal pipeline (Deploy website -> Docker -> GitHub -> Railway -> OBS notification).
- 3. Session Memory:
 - * *Lifespan*: Single calendar day (24 hours).
 - * *Examples*: Today's debugging log, morning meeting notes, study session summaries.
- 4. Project Memory:
 - * *Lifespan*: Project lifecycle (Weeks to months).
 - * *Examples*: Repository specs, design decisions, photography assets, YouTube video transcripts.
- 5. Long-Term Memory:
 - * *Lifespan*: Permanent / Continuous retention.
 - * *Examples*: User preferences, habits, personal goals, coding style rules, core relationships.
- 6. Knowledge Graph:
 - * *Lifespan*: Permanent graph nodes and semantic edges.
- 7. Obsidian Vault Mirror:
 - * *Lifespan*: Immutable human-readable .md vault files on local disk.

DOCUMENT 04.2 -- Knowledge Graph Engine

1. Graph Topology vs. Flat Folders

Traditional systems organize files into hierarchical directories. NAINA OS models knowledge as a Multi-Relational Directed Knowledge Graph, enabling deep associative reasoning across modalities.

```
graph TD
    Jonathan((Jonathan)) -->|MAINTAINS| NAINA_OS[Project: NAINA OS]
    Jonathan -->|STUDIES_AT| Aurora[Place: Aurora College]
    Jonathan -->|USES_DEVICE| SonyA58[Device: Sony A58 Camera]
    Jonathan -->|MANAGES| AcademicHub[Project: Academic Hub]

    NAINA_OS -->|CONTAINS_MODULE| Kernel[Subsystem: Microkernel]
    NAINA_OS -->|CONTAINS_MODULE| Voice[Subsystem: Voice Engine]
    NAINA_OS -->|CONTAINS_MODULE| Memory[Subsystem: Memory Engine]

    Memory -->|SYNC_TO_DISK| ObsidianVault[Vault: Local Markdown]
```

2. Schema for 16 Core Entity Types

The NKRS Knowledge Graph schema standardizes sixteen entity types and their allowed relationships:

```
// Knowledge Graph Entity Node Schema (TypeScript Contract)
export type EntityType =
  | 'Person' | 'Project' | 'Company' | 'Device'
  | 'File' | 'Note' | 'Task' | 'Goal'
  | 'Skill' | 'Event' | 'Place' | 'Photo'
  | 'Website' | 'AI_Model' | 'Agent' | 'Plugin';

export interface GraphNode {
  entityId: string; // UUIDv4 or URI
  entityType: EntityType;
  displayName: string;
  attributes: Record<string, unknown>;
  confidenceScore: number; // 0.0 to 1.0
  privacyLevel: 'PUBLIC' | 'PRIVATE' | 'RESTRICTED' | 'ENCRYPTED';
  createdAt: string; // ISO-8601
  lastAccessedAt: string;
}

export interface GraphEdge {
  edgeId: string;
  sourceEntityId: string;
  targetEntityId: string;
  relationshipType: string; // e.g., "CREATED_BY", "DEPENDS_ON", "LOCATED_AT"
  weight: number; // Relationship strength (0.0 to 1.0)
}
```

DOCUMENT 04.3 -- Obsidian Intelligence Framework

1. Vault Directory Architecture

Every note inside the local Obsidian Vault is an Intelligent Machine-Readable Object:

```
NainaSecondBrain/
+-- 00 Dashboard/      # System Status & Daily Summary Widgets
+-- 01 Daily Notes/    # Daily Session Journals (YYYY-MM-DD.md)
+-- 02 Projects/       # Active Project Hubs & Specifications
+-- 03 Research/       # Paper Summaries & Web Scrape Clippings
+-- 04 People/        # Contacts, Collaborators & Profiles
+-- 05 AI/            # Prompt Libraries, Model Benchmarks & Specs
+-- 06 Coding/        # Code Snippets, Architecture Specs & ADRs
+-- 07 College/       # Academic Notes, Syllabi & Assignment Logs
+-- 08 Photography/   # Camera Gear, EXIF Metadata & Shooting Logs
+-- 09 Journal/       # Personal Thoughts & Reflection Entries
+-- 10 Meetings/      # Meeting Transcripts & Action Items
+-- 11 Tasks/        # Atomic Task Items & Goal Trees
+-- 12 Media/        # Images, Diagrams & Voice Clips
+-- 13 Archives/     # Inactive Projects & Historical Logs
+-- 99 System/       # NAINA OS Metadata, Schemas & Indices
```

2. Standard YAML Frontmatter Schema

Every .md file generated or managed by NAINA OS incorporates strict metadata frontmatter:

```
---
title: "NAINA OS Kernel Specification"
created: "2026-07-31T21:42:41Z"
updated: "2026-07-31T21:56:44Z"
owner: "Jonathan"
agent: "CENANI"
project: "NAINA OS"
tags: ["engineering", "kernel", "microkernel", "specification"]
entities:
  - "Person:Jonathan"
  - "Project:NAINA OS"
  - "Subsystem:Microkernel"
embeddings:
  model: "text-embedding-3-small"
  dimensions: 1536
  vector_id: "vec_k8912a_kernel"
relationships:
  - relation: "MAINTAINED_BY"
    target: "Person:Jonathan"
  - relation: "PART_OF"
    target: "Project:NAINA OS"
status: "APPROVED"
---
```

DOCUMENT 04.4 -- Layered Memory Retrieval Pipeline

Rather than relying solely on naive vector embedding similarity, NAINA OS employs a 9-Stage Layered Retrieval Pipeline:

```
flowchart TD
    Q[User Prompt / Question] --> S1[Stage 1: Intent & Semantic Classification]
    S1 --> S2[Stage 2: Query Active Working Memory]
    S2 --> S3[Stage 3: Query Active Project Context]
    S3 --> S4[Stage 4: Knowledge Graph Traversal 2-hop]
    S4 --> S5[Stage 5: Obsidian Vault Frontmatter Lookup]
    S5 --> S6[Stage 6: Vector Similarity Search pgvector / ChromaDB]
    S6 --> S7[Stage 7: External Web Search if local confidence < 0.65]
    S7 --> S8[Stage 8: Context Assembly & Goal Planner Synthesis]
    S8 --> S9[Stage 9: Final Response Generation]
```

DOCUMENT 04.5 -- Memory Lifecycle Engine

Every memory object flows through nine lifecycle states:

```
[Created] --> [Indexed] --> [Embedded] --> [Linked] --> [Updated]
|
[Deleted] <-- [Recovered] <-- [Archived] <-----+----->
```

1. Created: Captured from voice, chat, vision, or file input.
2. Indexed: Parsed for entities, keywords, and metadata.
3. Embedded: Converted into dense vector space (1536-dim).
4. Linked: Attached to Knowledge Graph nodes and Obsidian frontmatter.
5. Updated: Modified as new context or facts emerge.
6. Archived: Shifted to cold storage when access frequency drops.

7. Recovered: Restored from archives upon relevant query match.
8. Deleted: Safely purged upon explicit user privacy request.

DOCUMENT 04.6 -- Memory Quality System

To reason accurately about its memories, NAINA OS tracks eight metadata quality dimensions for every memory record:

Quality Dimension	Type / Range	Purpose & Description
---	---	---
Confidence	Float (0.0 - 1.0)	Probability that the memory node is
Source	String Enum	Origin: VOICE, GUI_AUTOMATION, USER
Timestamp	ISO-8601 UTC	Exact creation time for temporal or
Importance	Integer (1 - 5)	Priority rating determining retenti
Privacy Level	String Enum	Access restriction: PUBLIC, PRIVATE
Related Entities	Array[EntityURI]	Direct graph edges linking to assoc
Verification Status	String Enum	Validation state: UNVERIFIED, USER_
Last Access	ISO-8601 UTC	Recency marker used for memory deca

SPECIAL SUBSYSTEM 1: Personal Knowledge Timeline (PKT)

The Personal Knowledge Timeline (PKT) provides a unified chronological view of user projects, activities, and achievements.

```
2026
+-- Jan 15: Started NAINA OS Design Specs
+-- Mar 10: Added OpenClaw Computer Use Integration
+-- May 20: Built Multi-Tier Memory Engine (MKIE)
+-- Jul 31: Released NAINA OS Specification v1.0
+-- Aug 15: Initiated ROS2 Robotics Module Integration
```

Interactive Drill-Down Capabilities

Clicking any timeline event immediately synthesizes and displays:

- * Associated Git commits and code diffs.
- * Relevant Obsidian notes and meeting logs.
- * Voice chat transcripts and planning contracts.
- * High-resolution screenshots, photos, and media captures.

SPECIAL SUBSYSTEM 2: Identity Engine

The Identity Engine serves as the master anchoring subsystem for NAINA OS, granting every memory, task, project, and permission a clear owner:

+-----+ 			-----+ 		
			IDENTITY ENGINE MATRIX		
+-----+ 			-----+ 		
User Profile			Master User Identity (Jonathan), Preferences, Keys		
+-----+ 			-----+ 		
Devices			Desktop Rig, Laptop, Android Phone, Sony A58 Camera		
+-----+ 			-----+ 		
Projects			NAINA OS, Academic Hub, Photography, YouTube Channel		
+-----+ 			-----+ 		
Organizations			Aurora College, Open Source AI Collective		
+-----+ 			-----+ 		
Collaborators			Friends, Co-developers, Research Partners		
+-----+ 			-----+ 		
Digital Identities			GitHub Handle, Google Workspace ID, OBS Socket Tokens		
+-----+ 			-----+ 		

End of NAINA OS Memory & Knowledge Intelligence Engine -- Documents 04.1 through 04.6, PKT & Identity Engine